3 Sum

Hectron

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1 Description

Given an array nums of n integers, return an array of **all the unique quadruplets**: [nums[a], nums[b], nums[c]] such that:

- 0 <= a, b, c, d < n
- a, b, c, and d are distinct.
- nums[a] + nums[b] + nums[c] + nums[d] == target

2 Function/Method Declaration

std::vector< std::vector<int> > threeSums(std::vector **numbers**, int target);

3 Solution

I. Mix **numbers** with itself to find all pair combinations.

II. Recent sum pairs produced are crossed with **numbers** to produce another list of pairs.

III. II is repeated N-2 times (i.e. N=4; 2 iterations)

3.1 First pairs

Method below is used to find unique pairs of two equal vectors (i.e. autocorrelation sampless).

Raw vector: numbers = [0, -4, 3, 1, 0]

Sort set: numbers = [-4, 0, 0, 3, 3]

Unqiue values in vector: numbers = [-4, 0, 1, 3]

Pairs (diagonal search):

Table 1: Sums

List	-4	0	1	3
-4	**-8**	-4	-3	-1
0	**-4**	**0**	1	3
1	**3**	**1**	**2**	4
3	**-1**	**3**	**4**	**3**

3.2 Other pairs

Method below is used to find unique pairs of input vector with previous sum pairs:

Input set: numbers = [-4, 0, 1, 3]

Sums set: numbers = [-8, -4, -1, 0, 1, 2, 3, 4]

Table 2: Sums

List	-4	0	1	3
-8	**-12**	-8	-7	-5
-4	**-8**	**-4**	-3	-1
-1	**-5**	**-1**	**0**	2
0	**-4**	**0**	**1**	**3**
1	**-3**	**1**	**2**	**4**
2	**-2**	**2**	**3**	**5**
3	**-1**	**3**	**4**	**6**
4	**0**	**4**	**5**	**7**

3.3 FlowChart